

FakeDetect

AI-Powered Multimodal Fake News & Deepfake Detection System



Problem Statement



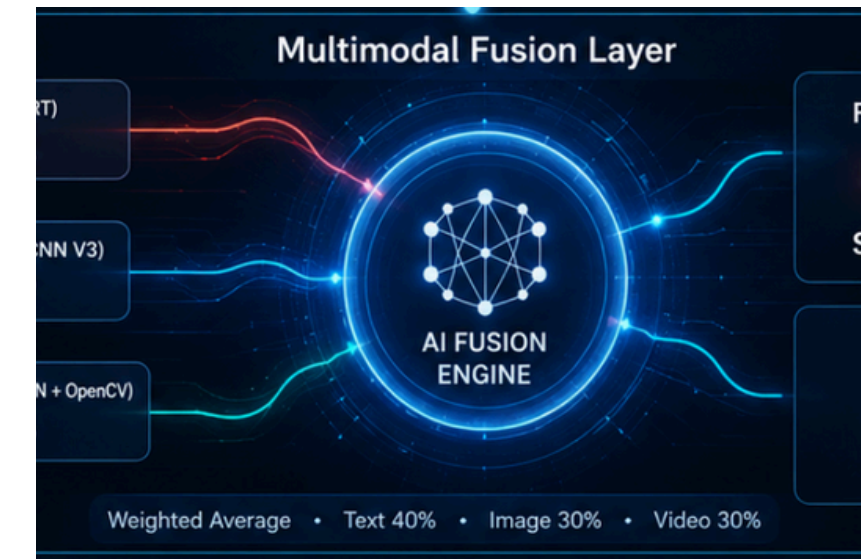
Fake News is Rampant

False and misleading news spreads rapidly across digital platforms, influencing public opinion, shaping political narratives, and making it difficult to distinguish credible information from misinformation.



Deepfakes are Becoming More Realistic

AI-generated images and videos can convincingly imitate real people and events. Modern deep learning techniques produce highly realistic synthetic media that is increasingly difficult to detect.



The Need for a Multimodal Solution

real-world misinformation often combines fake news with manipulated images or videos, there is a need for a unified multimodal detection system capable of analyzing all three modalities together.

Objectives

01

Develop a multimodal AI system to detect fake news and deepfake content from **text, images, and videos**.

02

Classify fake news using a fine-tuned **BERT** model.

03

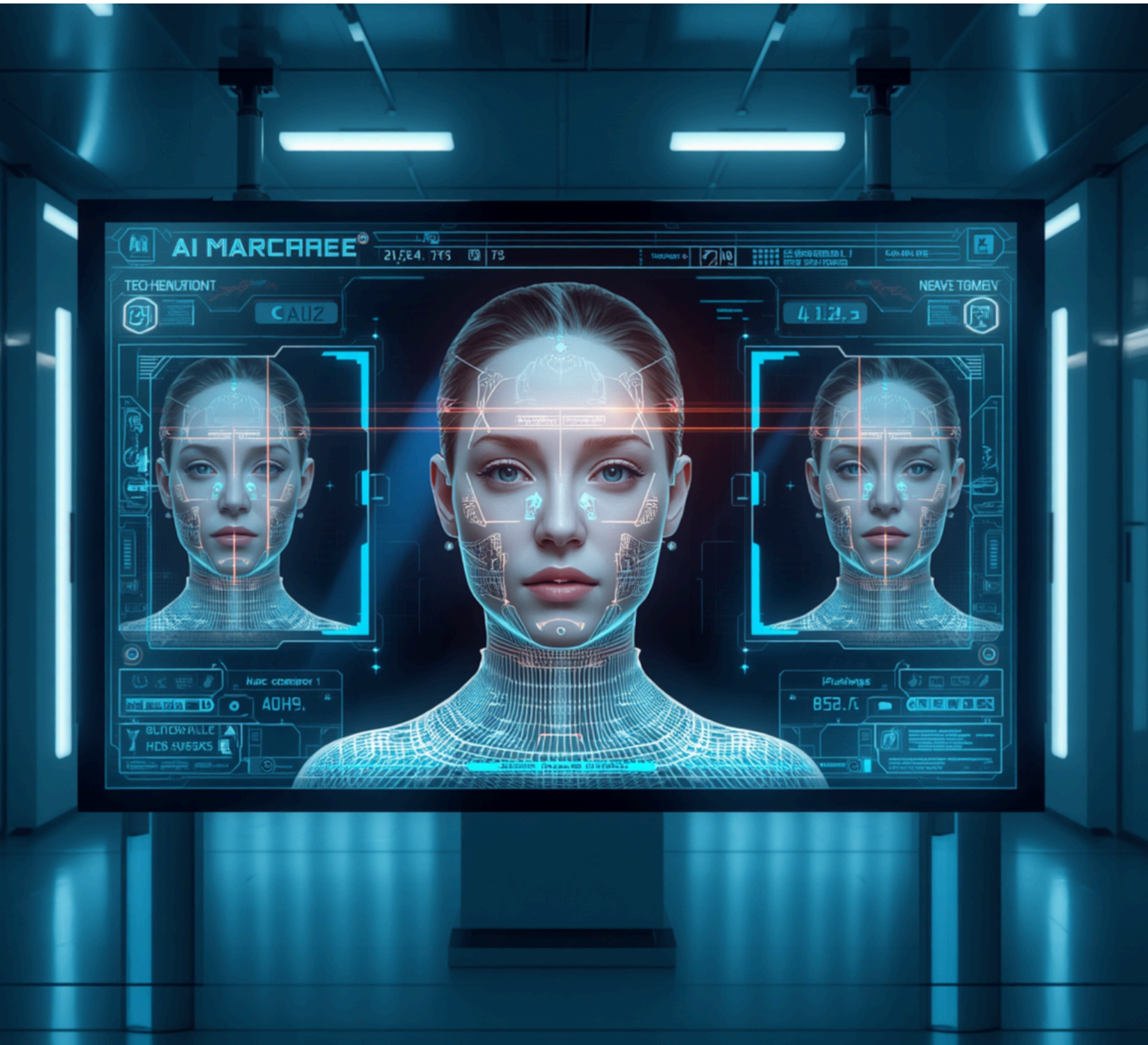
Detect manipulated images and videos using **EfficientNetB0 V3**.

04

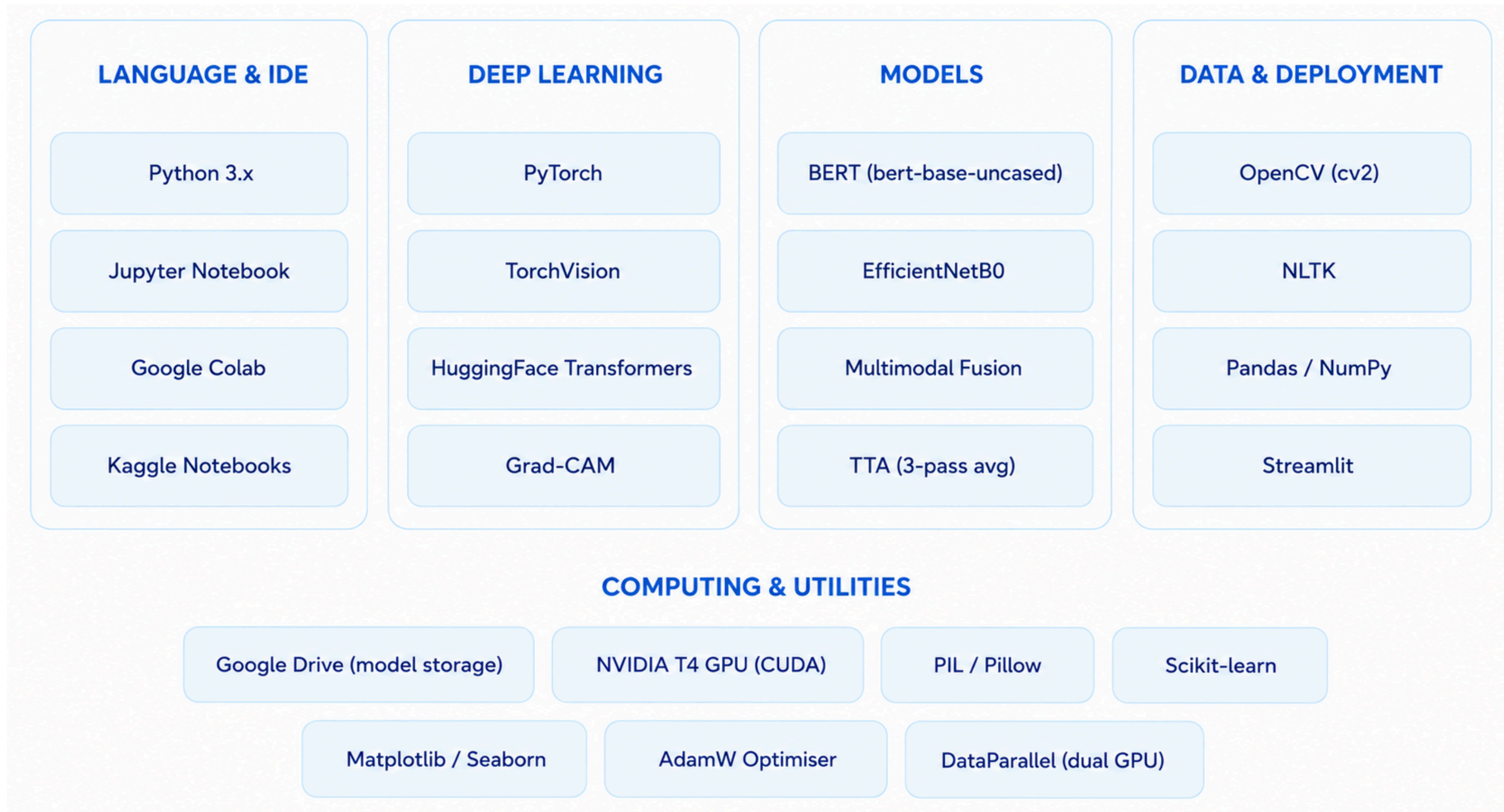
Combine text, image, and video predictions through **multimodal fusion** for improved accuracy.

05

Deploy the system as a **Streamlit web application** for real-time fake content detection.



Tools and Technologies Used



System Overview



FakeDetect

FakeDetect is a multimodal AI-powered system designed to detect fake news and deepfake content from text, images, and videos. It combines Natural Language Processing (NLP) and Computer Vision techniques to analyze different types of digital media within a single platform. The system provides accurate predictions along with visual explanations, making the detection process transparent and reliable.

BERT

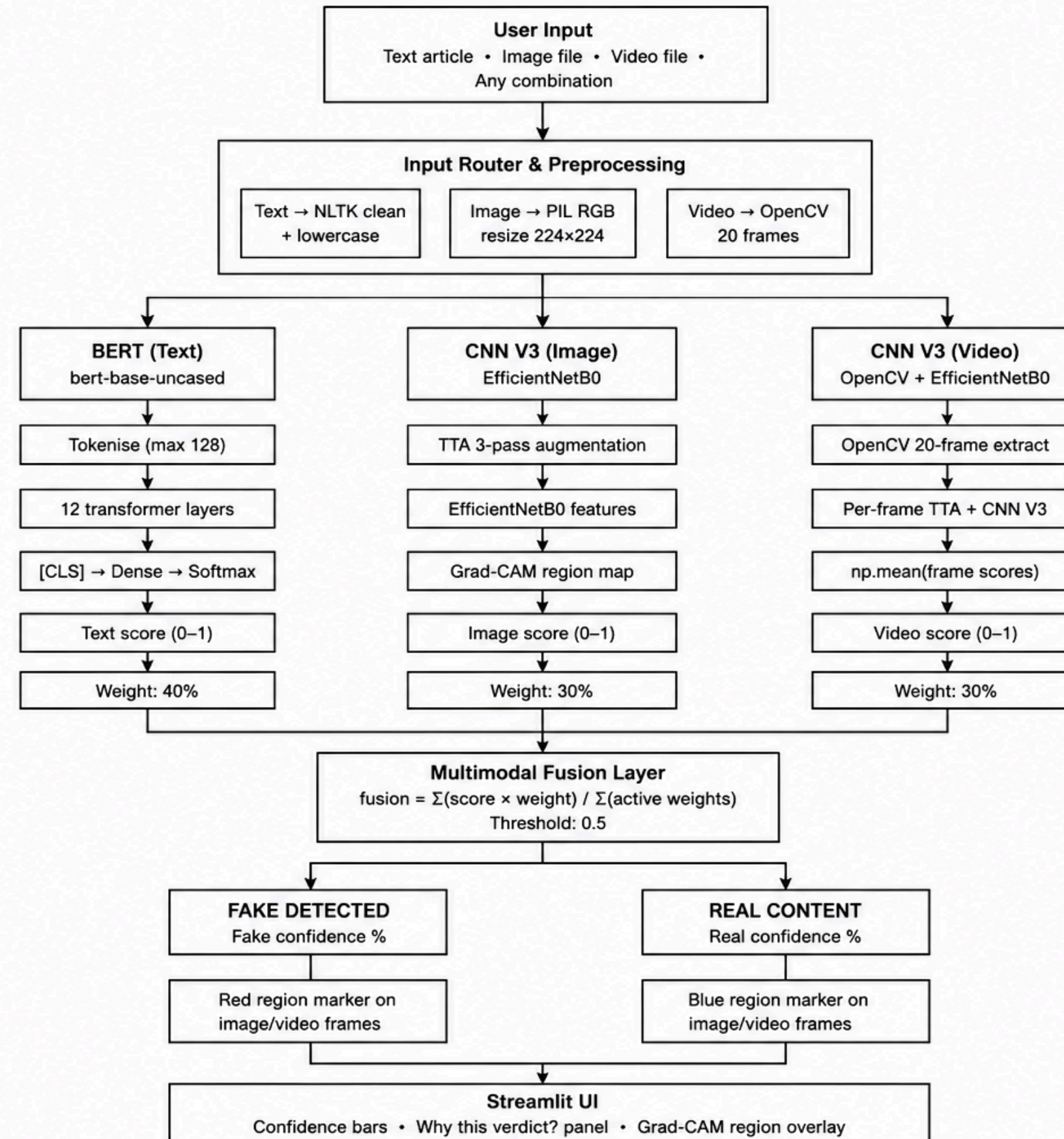
Detects fake news by understanding the contextual meaning of news articles using the BERT transformer model. It classifies text as Real or Fake with high accuracy

EfficientNetB0

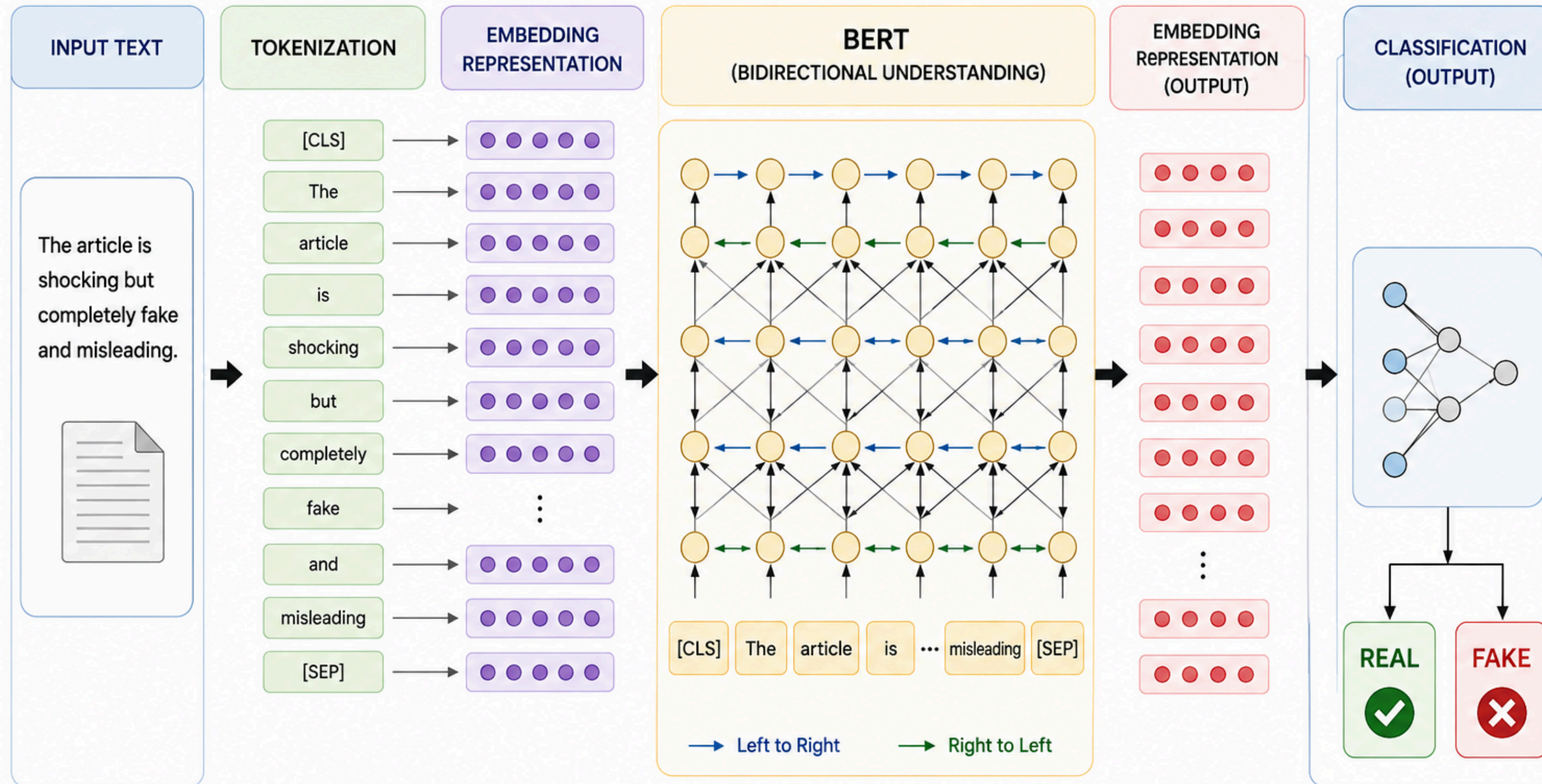
Uses EfficientNetB0 to identify manipulated images and video frames. Grad-CAM highlights the regions that influenced the model's prediction, improving explainability.

Multimodal Fusion

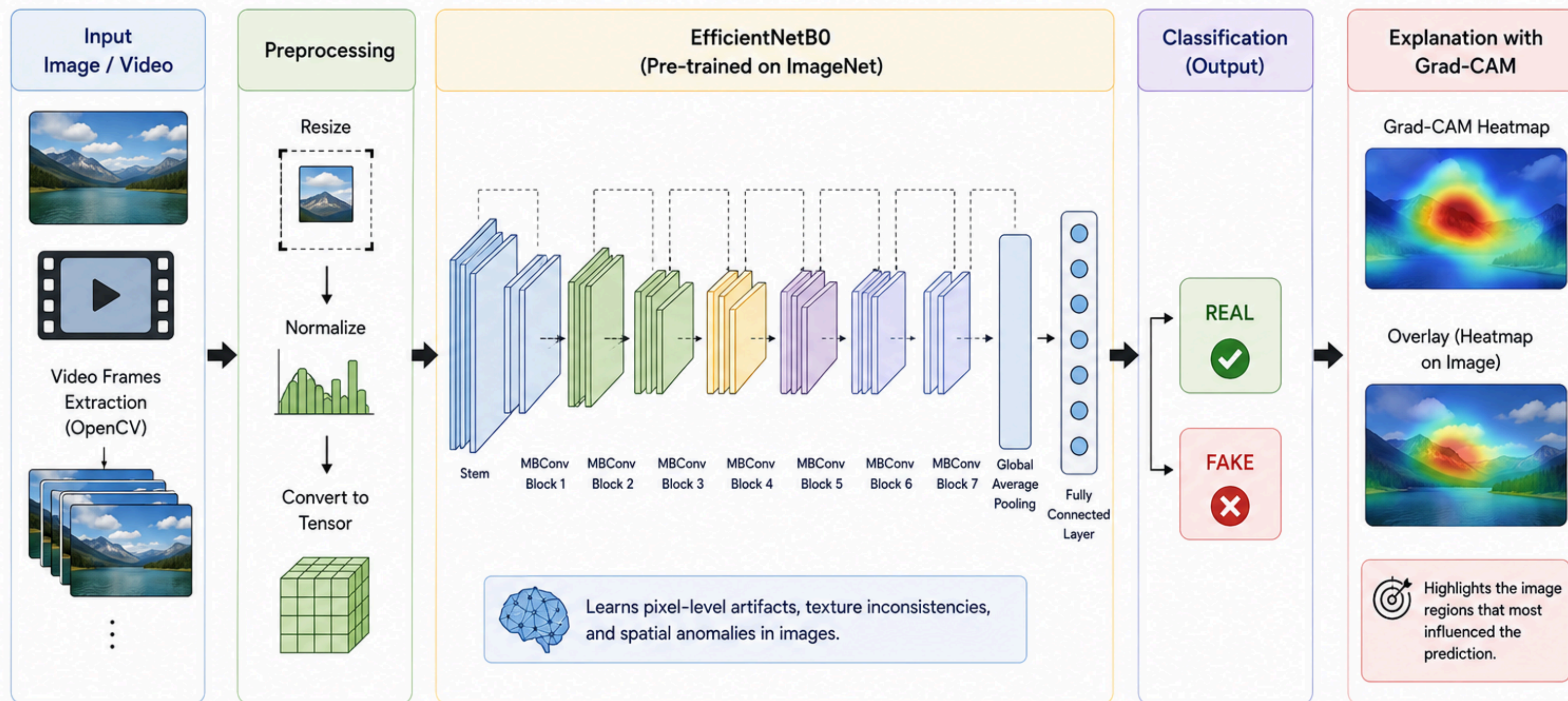
Combines predictions from text, image, and video models using a weighted fusion strategy to produce a single, reliable final decision. The system is deployed through Streamlit for real-time user interaction.



BERT WORKFLOW



EfficientNetB0 Workflow



Datasets & Training

6 datasets · 4,36,000+ training samples across all modules

Dataset	Type	Size	Purpose
Fake and Real News (Kaggle)	Text CSV	44,898 articles	BERT training
LIAR Dataset (Kaggle)	Text CSV	12,836 statements	BERT secondary
Deepfake vs Real 60K	Images	190,335 images	CNN face-swap training
140K Real & Fake Faces	Images	140,000 images	CNN StyleGAN faces
AI-Generated vs Real	Images	48,000 images	CNN AI-generated content
Real/Fake Video Dataset	Videos	872 clips	Video fine-tuning

99.13%

BERT
accuracy

92.34%

Image
accuracy

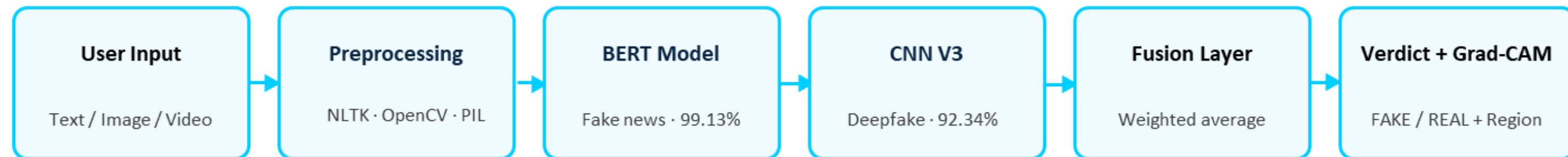
99.89%

Video
accuracy

92.34%

Fusion
accuracy

System Architecture



Text Pipeline

- Text Cleaning & Lowercasing
- NLTK Tokenization
- BERT Tokenizer (128 Tokens)
- BERT Classification
- Fake News Prediction

Image/Video Pipeline

- Image Resize (224 × 224)
- Image Normalization
- TTA (3-Pass Augmentation)
- EfficientNetB0 V3 Inference
- Grad-CAM Visualization

Fusion Strategy

- Text Weight: 40%
- Image Weight: 30%
- Video Weight: 30%
- Auto Weight Normalization
- Final Multimodal Prediction

Results & Evaluation

BERT (Text)

Training: 44,898 articles

99.13%

Accuracy

99.94%

Precision

98.40%

Recall

99.16%

F1-Score

CNN V3 (Image)

Training: 288K+ images

92.34%

Accuracy

91.32%

Precision

93.58%

Recall

92.44%

F1-Score

CNN (Video)

Training: 872 video clips

99.89%

Accuracy

99.75%

Precision

100.0%

Recall

99.88%

F1-Score

Fusion System

Training: Combined

92.34%

Accuracy

91.32%

Precision

93.58%

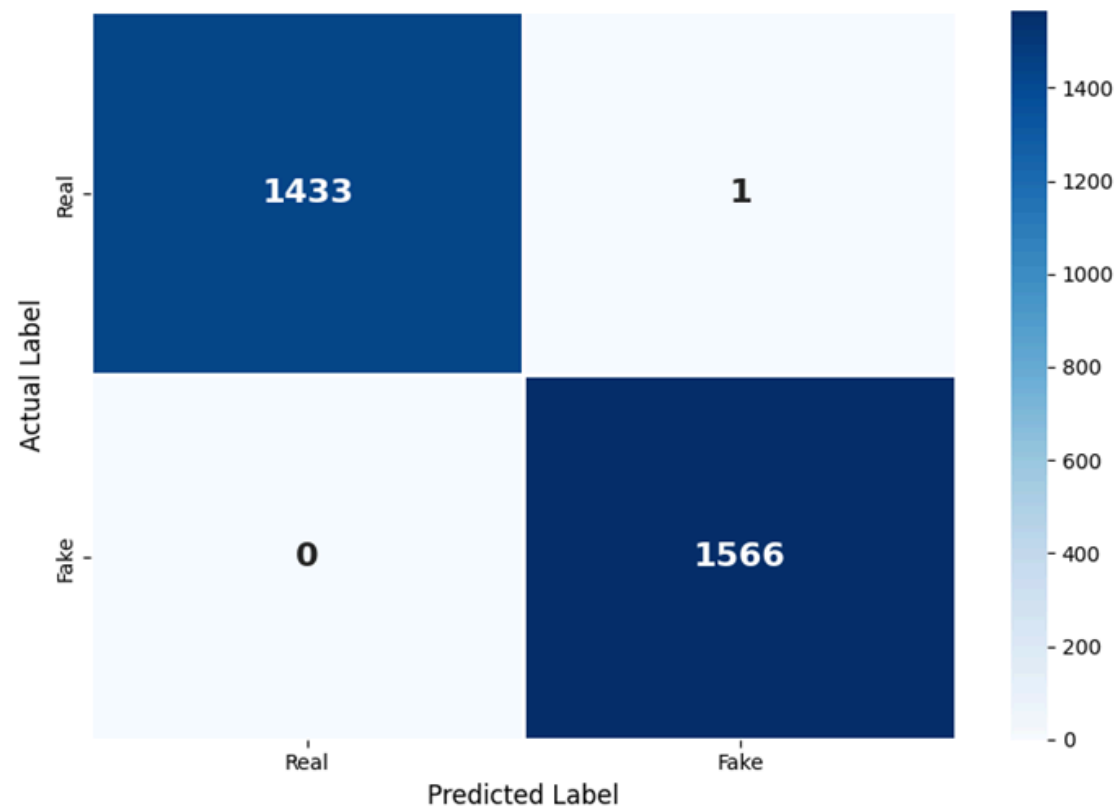
Recall

92.44%

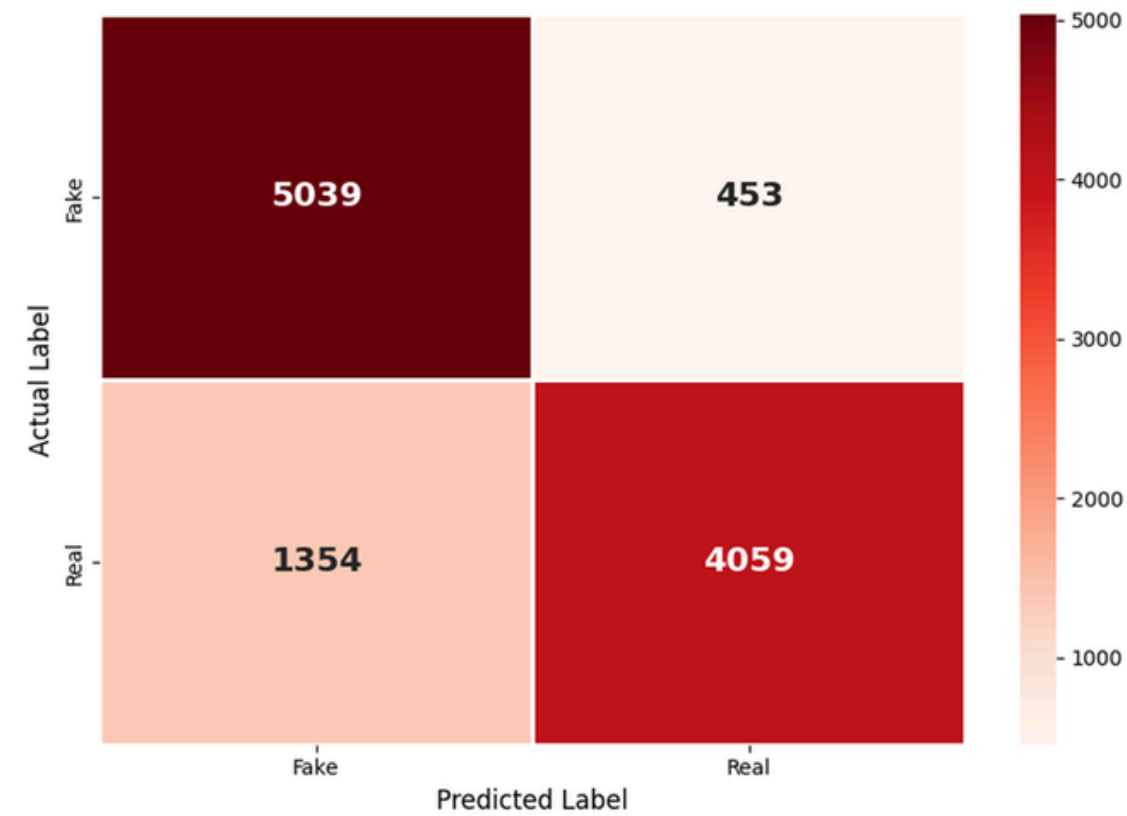
F1-Score

Model Performance Visualization

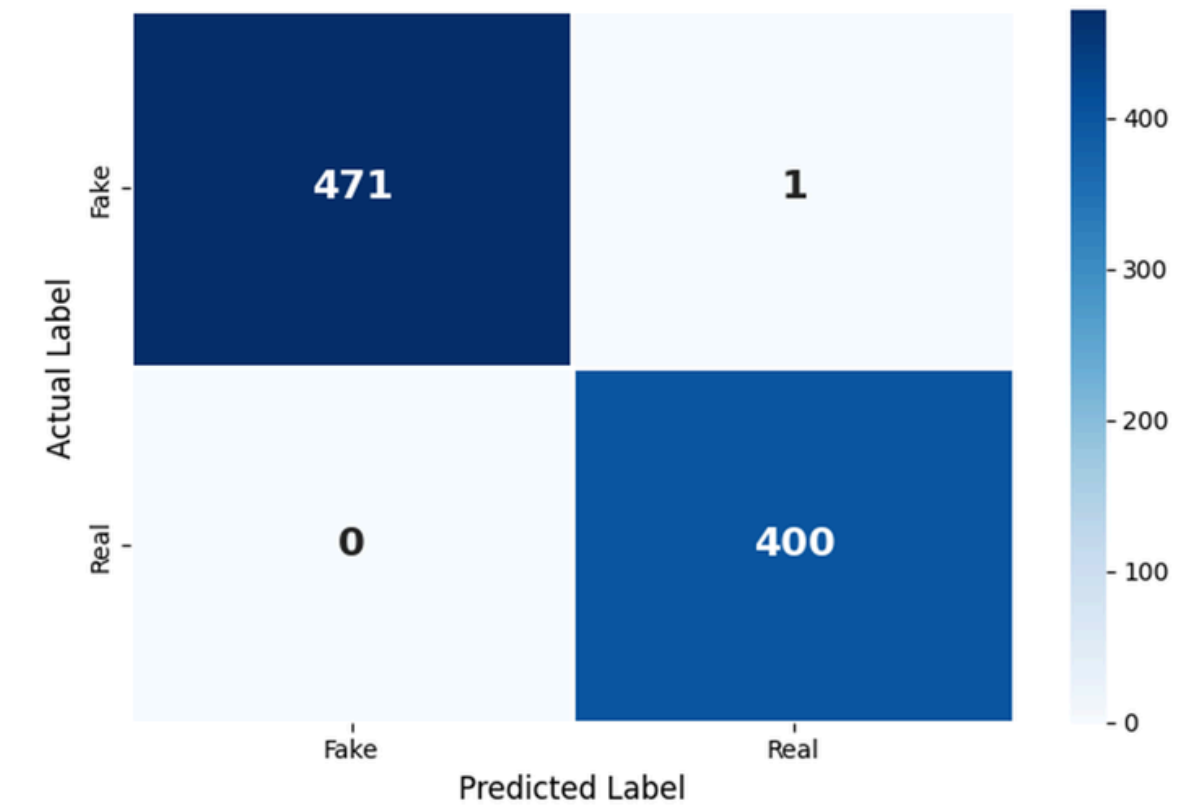
Confusion Matrix — BERT Fake News Detection



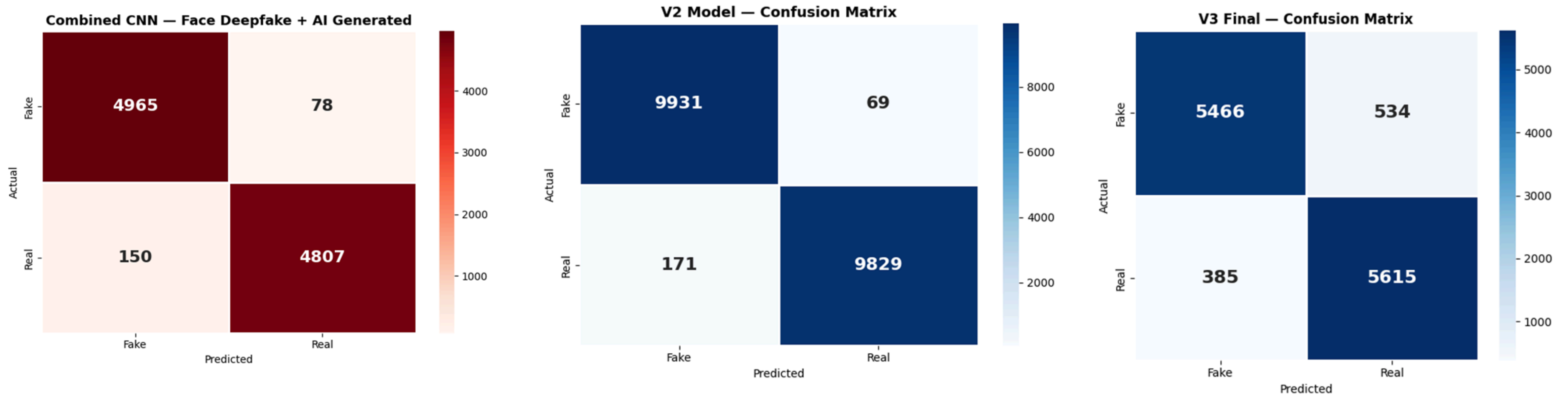
Confusion Matrix — EfficientNetB0 Deepfake Detection



Confusion Matrix — Fine-tuned Video Detection



Model Performance Visualization



GRAD-CAM VISUALS

00JEP4Z36Z.jpg | Real: 0.0% Fake: 100.0%

Input

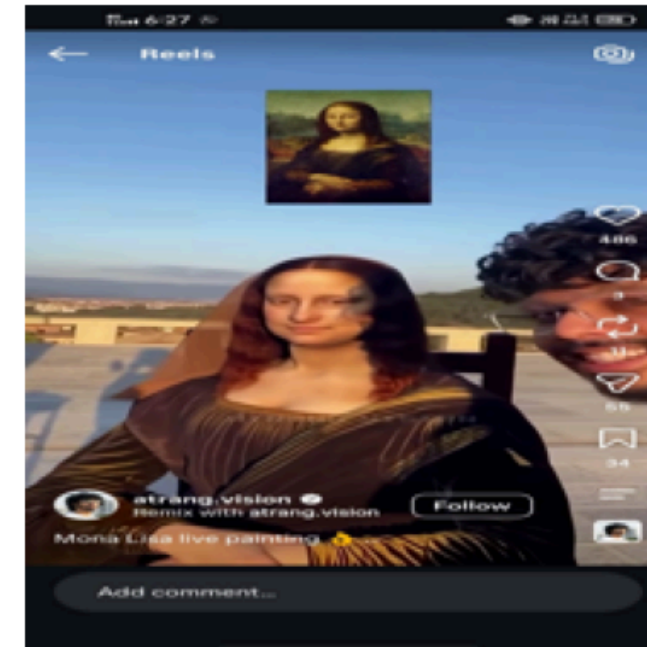


Grad-CAM — FAKE

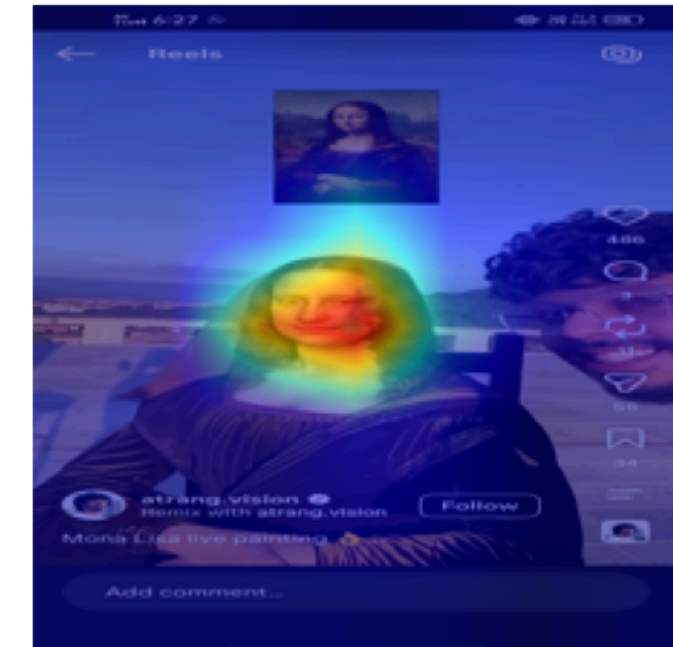


WhatsApp Image 2026-07-07 at 6.41.28 PM.jpeg | Real: 32.1% Fake: 67.9%

Input



Grad-CAM — FAKE



00PYA83V1P.jpg | Real: 0.0% Fake: 100.0%

Input

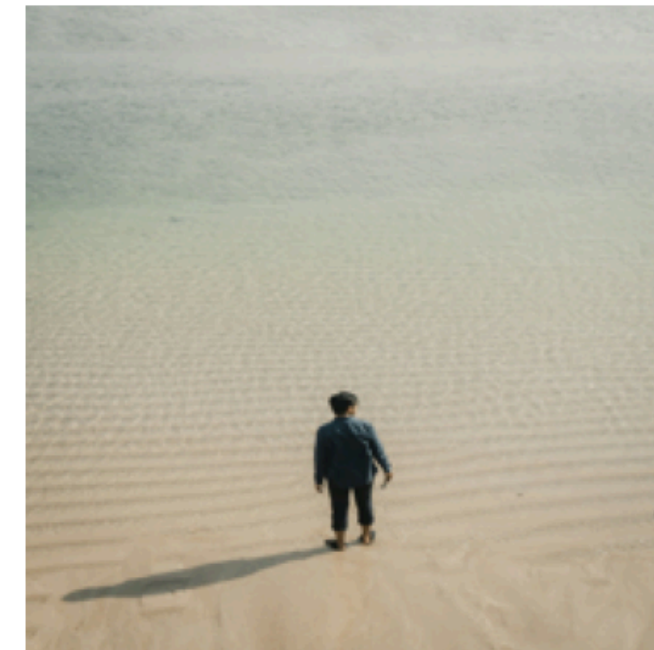


Grad-CAM — FAKE

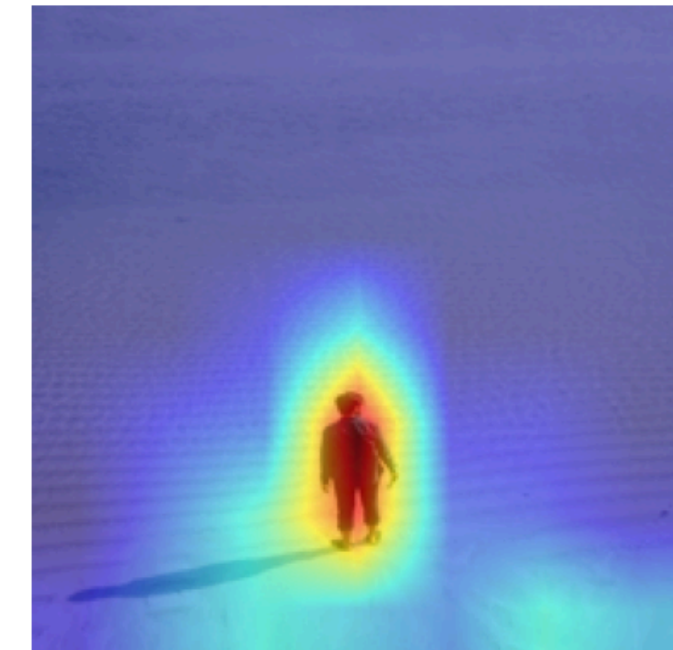


0011.jpg | Real: 71.8% Fake: 28.2%

Input



Grad-CAM — REAL

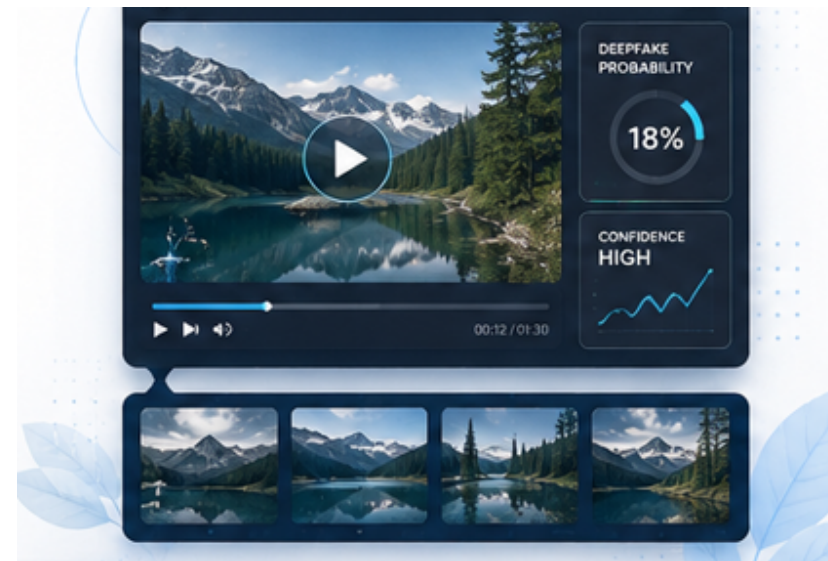


Conclusion



Multimodal Detection System

Developed FakeDetect, an AI-driven multimodal system capable of detecting fake news and deepfake content from text, images, and videos.



High Detection Performance

Achieved high detection performance by combining BERT, EfficientNetB0, and multimodal fusion, with Grad-CAM providing visual explainability.



Real-Time Deployment

Successfully deployed the system as a Streamlit web application for real-time, user-friendly fake content detection.

Thank You

